Tribhuvan University

Institute of Science and Technology



Central Department of Computer Science and Information Technology

Kirtipur, Kathmandu

A Thesis Proposal On

Regime-Aware Stock Movement Prediction in NEPSE Using Hidden Markov Models and Gradient Boosting

Under the supervision of Asst. Prof. Jagdish Bhatta

In partial fulfillment of the requirements for the Master's degree in Information Technology (MIT)

Submitted to:

Central Department of Computer Science and Information Technology,

Tribhuvan University, Kirtipur, Kathmandu, Nepal

Submitted By:

Manish Shrestha (7925011)

(5-1-282-466-2007)

2026

Table of Contents

List of Figures iii
List of Tables iv
List of Abbreviations v
CHAPTER 1: INTRODUCTION 1
1.1 Introduction 1
1.2 Problem Statement 2
1.3 Objectives 3
CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW 4
2.1 Background Study 4
2.2 Literature Review 5
CHAPTER 3: METHODOLOGY 8
3.1 Dataset Description 8
3.2 Data Preprocessing 10
3.2.1 Data cleaning and standardization 10
3.2.2 Feature Engineering 10
3.2.3 Target Construction 10
3.2.4 Chronological Data Partitioning 11
3.3 Implementation Model 11
3.3.1 Gaussian Hidden Markov Model for Regime Detection 11
3.3.2 Gradient Boosting Models for Prediction 15
3.4 Evaluation Metrics 18
3.4.1 Macro F1 Score 19
3.4.2 Cumulative Return 19
3.4.3 Sharpe Ratio 19
3.4.4 Maximum Drawdown 19
3.4.5 Win Rate 20
CHAPTER 4: EXPECTED OUTCOMES AND WORK SCHEDULE 21
4.1 Expected Outcomes 21
4.2 Work Schedule 21
REFERENCES 22

List of Figures

Figure 3.1: System Flowchart 8

Figure 3.2: 3-state Gaussian HMM Architecture 12

Figure 3.3: LightGBM Architecture 16

Figure 3.4: XGBoost Architecture 17

Figure 4.1: Working Schedule 21

List of Tables

Table 3.1 Selected NEPSE scrips used in the study 9

List of Abbreviations

ADX	Average Directional Index
ATR	Average True Range
HMM	Hidden Markov Model
LightGBM	Light Gradient Boosting Machine
MACD	Moving Average Convergence Divergence
MDD	Maximum Drawdown
ML	Machine Learning
NEPSE	Nepal Stock Exchange
OHLCV	Open High Low Close Volume
RSI	Relative Strength Index
XGBoost	eXtreme Gradient Boosting

CHAPTER 1: INTRODUCTION

1.1 Introduction

Forecasting stock market movement is tough because financial time series data are noisy, non-linear and non-stationary which is also influenced by investor sentiment against changing market scenario. Most of the markets' statistical properties of returns are not stable over time, meaning the market faces different hidden states from slow and steady to paced and turbulent periods. This makes regime-aware modeling vital because a prediction model trained under one market condition may or may not perform as same under another. Hidden Markov Models (HMMs) are widely used for this purpose because they can infer unobserved market states from observable variables such as returns and volatility, while recent financial forecasting studies continue to combine regime detection with machine learning (ML) to improve predictive robustness as researched under the paper of (Okşak, Büyükkör, & Sarıtaş, 2026).

At the same time, ML models, especially gradient boosting methods, have become strong tools for stock movement prediction when the problem is expressed through engineered tabular features such as returns, volatility measures, trend indicators, and market-relative measures. Compared with many deep learning approaches, gradient boosting models are often more practical in settings where the dataset is moderate in size and features are carefully prepared. Recent literature in financial forecasting and regime-aware prediction continues to show strong use of ML and hybrid market-state approaches (Pagliaro, 2026).

The Nepal Stock Exchange (NEPSE) provides a meaningful background for such a study. NEPSE is the only stock exchange in Nepal and contains a market structure heavily influenced by financial institutions and operators, while the market reflects the characteristics of an emerging market, including limited depth, changing volatility, and cross-sectional heterogeneity across listed companies. These features make NEPSE a suitable environment for examining whether regime-aware methods can improve stock movement prediction beyond static feature-based models.

This study seeks to address these challenges by preparing a regime-aware ML framework for stock movement prediction in NEPSE using HMM and gradient boosting. The study uses historical Open, High, Low, Close, Volume (OHLCV) data from selected NEPSE-listed scrips together with NEPSE index data. A market-level

HMM will be used to detect overall volatility regimes, while separate stock-level HMMs will be used to detect scrip-specific regimes. These regime features will be then incorporated into gradient boosting models for Buy, Hold, and Sell prediction. In addition to predictive evaluation, the study will also consider the practical usefulness of the generated signals through an out-of-sample backtesting framework. This will allow the study to evaluate both statistical prediction quality and economic relevance in the NEPSE context.

1.2 Problem Statement

Most stock prediction studies assume that the relationship between input features and future price remains relatively stable throughout the time but in practice, stock market often pass through different latent conditions characterized by changing volatility, uncertainty and return behavior. When such market state changes are not considered, a single prediction model may not fully able to capture the different phases of market's lifecycle like accumulation, advancing, distribution and declining which results in poorer generalization. As an emerging market, NEPSE is likely to show changing volatility and structural variation across scrips and sectors, yet there is limited evidence on whether market-regime information can be systematically combined into stock movement prediction for this market. There remains a gap in applying a leakage-aware regime-aware prediction framework tailored to NEPSE using both market-level and stock-level volatility states. The problem is therefore not only whether stock movement can be predicted, but whether combining volatility regimes improves Buy/Hold/Sell prediction performance and practical signal usefulness in NEPSE. Further, in time series prediction, improper data splitting or full sample regime fitting can introduce look-ahead bias, causing the results of the prediction model to be too accurate. For regime-aware financial modeling, this issue is concerning because latent-state detection itself can accidentally use future information if it is not handled carefully. Hence, a proper time-based train, validation, and test design is vital to assess whether regime-aware prediction improves out-of-sample performance.

This highlights the need for regime-aware methods that can detect latent market states based on volatility levels which can improve the stock movement prediction in NEPSE evaluated using a leakage-aware gradient boosting framework.

1.3 Objectives

The objectives of this study are:

1. To implement and analyze a regime-aware gradient boosting framework for Buy, Hold, and Sell stock movement prediction in NEPSE using HMM for volatility regime detection.
2. To evaluate the trading-oriented performance of the proposed framework using a simplified out-of-sample backtesting design.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

The various properties of financial time-series such as noise, non-linearity, non-stationary, structural breaks, volatility clustering, changing market conditions, and investor and trader sentiments make it difficult for its forecasting since models that assume only stable data generation often struggle to capture those patterns. This is why regime-aware modeling has become important in modern financial forecasting.

A key concept behind regime-aware forecasting is that markets move through hidden states such as low, moderate or high volatility periods. HMMs are widely used to model such latent regimes because they can infer unobserved states from observable variables such as returns, volatility, and momentum-related measures. Recent work proposed by (Koukorinis, Germano, & Peters, 2025) continues to use HMMs either directly for regime identification or as part of hybrid ML pipelines, especially where the goal is to improve robustness across different market conditions.

Despite the availability of many deep learning architectures that work well in advanced stock markets of the developed countries, gradient boosting models such as XGBoost and LightGBM have become strong choices for financial prediction when the data are represented as engineered tabular features, and are moderate in size, the features are heterogeneous, and class or signal level interpretation is still important, which is relevant for emerging markets like NEPSE, where data are often smaller, noisier, and less standardized than in major developed markets. NEPSE being the sole stock exchange in Nepal reflects the characteristics of an emerging market, including sector concentration, varying liquidity, and strong shifts in volatility across time. Recent NEPSE-focused studies show growing interest in AI-based prediction, but the literature is still relatively limited and is dominated by deep learning as proposed in (Pokharel, et al., 2022), sentiment-based prediction as proposed in (Dahal, Gupta, & Pokharel, 2024), and conventional machine learning rather than leakage-aware regime-aware frameworks.

For this reason, a regime-aware prediction framework is particularly relevant to NEPSE. If market and stock behavior differ across hidden volatility states, then including those states as explanatory features may improve Buy/Hold/Sell prediction

compared with static feature-only models. The present study builds on this insight by combining HMM-based regime detection with gradient boosting classification under a time-based, leakage-aware evaluation design.

2.2 Literature Review

The start of this study begins with the broad stock-prediction literature. (Olorunnimbe & Viktor, 2023) surveyed deep learning applications in stock markets and found that prediction studies increasingly emphasize not only predictive accuracy but also backtesting, reproducibility, and application realism. Their survey is important because it highlights a key issue for this thesis: performance claims in financial forecasting are more credible when they go beyond pure error metrics and include practice-oriented evaluation.

(Thakkar & Chaudhari, 2021) reviewed deep neural networks for stock prediction and showed that CNNs, RNNs, LSTMs, GRUs, and hybrid architectures have all been applied successfully, but they also pointed out persistent challenges such as overfitting, instability across datasets, and sensitivity to feature design and market condition. This supports the argument that more complex models are not automatically superior, especially in smaller or more volatile markets like NEPSE, an emerging stock market.

(Bisdoulis, 2025) investigated asset forecasting using LightGBM with feature engineering and transformation methods. The study emphasized that although hybrid deep learning and machine learning models can be effective, its computational requirements may be high. To address this limitation, the author optimized LightGBM as a computationally efficient gradient boosting approach for financial forecasting. The study introduced engineered features and tested different target transformations such as returns, logarithmic returns, and EMA-based ratios. The findings support the use of LightGBM in stock forecasting tasks where structured financial indicators are available and computational efficiency is important. Similarly, (Pabuccu & Barbu, 2024) proposed a feature selection-based approach for forecasting financial time series using logistic regression, XGBoost and LSTM models across multiple stock and cryptocurrency datasets, where it showed that feature selection improved model performance for both regression and classification tasks involving returns and return movements. The inclusion of XGBoost alongside LSTM is important because it shows that gradient boosting models remain competitive alternatives to deep learning models,

especially when technical indicators and selected financial features are used. This shows the gradient boosting models like LightGBM and XGBoost are well suited to medium-sized, feature-engineered tabular financial datasets such as the NEPSE dataset used in this study.

(Koukorinis, Germano, & Peters, 2025) proposed a hybrid HMM-SVM/MKL framework for high-frequency financial regime classification. Their work is important because it demonstrates a general principle that is highly relevant to this thesis: HMMs can be used not only as standalone regime detectors but also as feature-generating components within a broader discriminative learning architecture. In their case, HMM-derived embeddings improved regime classification and trade-direction prediction.

(Kijkarncharoensin, 2025) used a multivariate hidden Markov framework to identify risk regimes in a sectoral stock index and showed that latent-state modeling can reveal structures not visible under conventional approaches. This is directly relevant to the present study because it supports the use of HMMs for volatility-state discovery in equity stock data and strengthens the methodological case for using interpretable hidden regimes such as low-volatility, medium-volatility, and high-volatility states.

Recent regime-aware work has gone further by explicitly combining hidden-state modeling with predictive systems. (Fons, Dawson, Yau, Zeng, & Keane, 2021) developed a dynamic asset allocation framework using Feature Saliency Hidden Markov Models and showed that HMM-based regime detection can improve risk-adjusted portfolio behavior by adapting to market cycles. Although their work is allocation-focused rather than classification-focused, it strongly supports the use of HMMs for identifying market cycles before making downstream decisions.

While studying many literatures based on NEPSE context, it suggests three things: first, there is clear local interest in applying ML and deep learning in stock prediction. Second, most NEPSE studies price forecasting or direction prediction rather than regime-aware classification. (Pokharel, et al., 2022) performed a comparative study of LSTM, GRU, and CNN for next-day NEPSE index prediction using fundamental, macroeconomic, technical, and financial text variables. They reported that LSTM achieved the strongest fit among the tested deep models. Third, a clearly peer-reviewed, recent NEPSE-specific HMM-based stock movement prediction study in the retrieved literature could not be verified. A NEPSE related work appears closer to Markov Chain

analysis than to a modern HMM-plus-ML prediction framework, where (Bhusal, 2017) applied a Markov Chain Model to the NEPSE by defining states as directly observable, daily price changes (Increase, Decrease, Same). This differs from an HMM, which assumes states (like returns, volatility, momentum) are latent and must be inferred from observations rather than directly calculated. This gap provides a strong justification for the present study.

CHAPTER 3: METHODOLOGY

The following diagram shows the overall system flowchart.

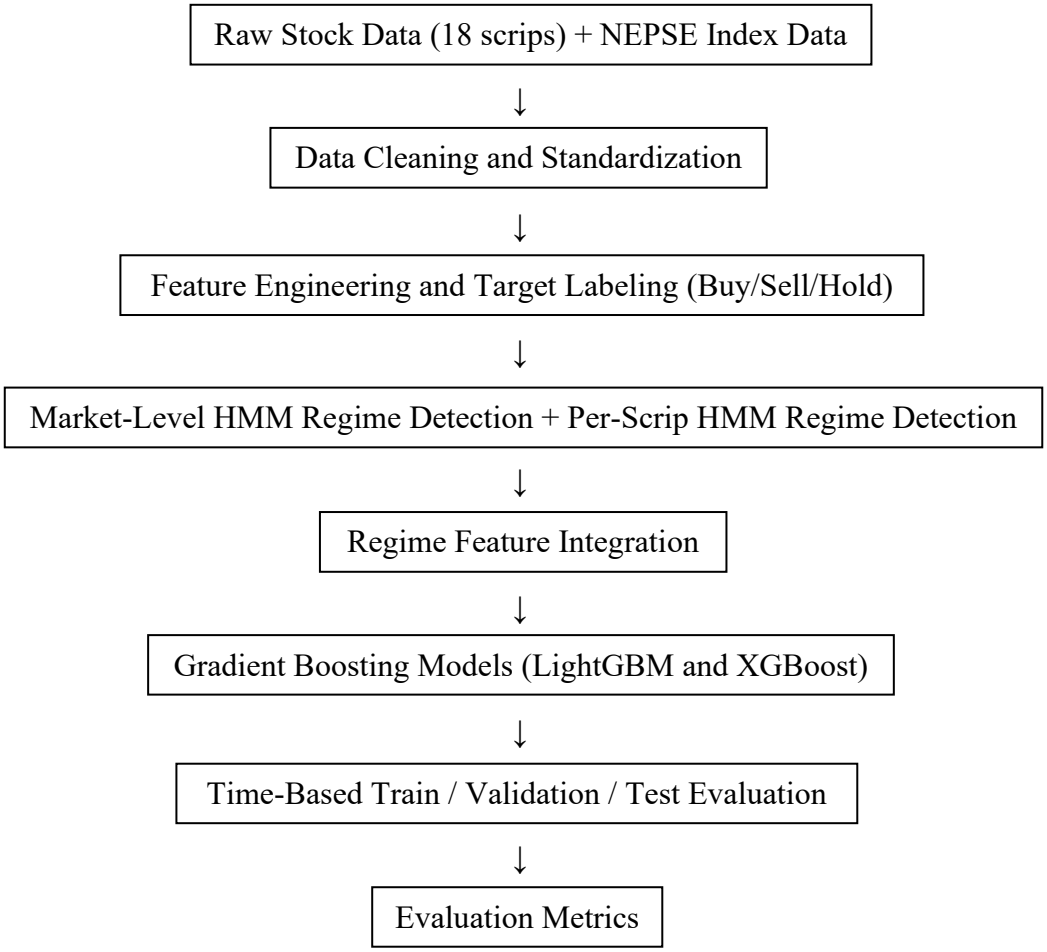


Figure 3.1: System Flowchart

It outlines a system that leverages the combination of HMM and gradient boosting models to predict the stock movement. The HMM detects the latent states of the market level as well as per scrip for all the 18 scrips which will be then integrated as the features to train the gradient boosting models.

3.1 Dataset Description

The dataset for this research will be constructed from historical daily OHLCV price adjusted data from 18 actively traded NEPSE-listed scrips representing multiple sectors with daily NEPSE index data obtained from ShareSansar, an online financial portal of Nepal (ShareSansar, n.d.). The study period spans from 2011 to January 2026, resulting in 52,551 observations after combining all selected scrips.

The dataset contains three categories of information:

1. Primary Data: Daily adjusted OHLCV records of the selected NEPSE-listed companies
2. Secondary Data: Daily NEPSE index data used to represent market-wide movements and conditions.
3. Derived Data: Features generated through statistical calculations and technical analysis, including returns, volatility measures, momentum indicators, volume indicators, volatility regime labels, and target labels.

The selected companies and their respective sectors are presented in Table 3.1.

Table 3.1 Selected NEPSE scrips used in the study

Sector	Scrip (Ticker)	Count
Hydropower	Butwal Power Company Limted (BPCL) Chilime Hydropower Company Limited (CHCL) Upper Tamakoshi Hydropower Limited (UPPER)	3
Microfinance	Chhimek Laghubitta Bittiya Sanstha Limited (CBBL) Deprosc Laghubitta Bittiya Sanstha Limited (DDBL)	2
Investment	Citizen Investment Trust (CIT)	1
Development Bank	Garima Bikas Bank Limited (GBBL) Muktinath Bikas Bank Limited (MNBBL)	2
Commercial Bank	Global IME Bank Limited (GBIME) Nabil Bank Limited (NABIL)	2
Finance	ICFC Finance Limited (ICFC)	1
Non-Life Insurance	NLG Insurance Company Limited (NLG) Shikhar Insurance Company Limited (SICL)	2
Life Insurance	Nepal Life Insurance Company Limited (NLIC) National Life Insurance Company Limited (NLICL)	2
Others	Nepal Telecom (NTC)	1
Hotel and Tourism	Taragaon Regency Hotel Limited (TRH) Soaltee Hotel Limited (SHL)	2
	Total Count	**18**

The use of multiple sectors will allow the proposed framework to capture heterogeneous market behavior and improve the generalizability of the predictive models across different sectors of NEPSE.

3.2 Data Preprocessing

3.2.1 Data cleaning and standardization

All datasets will be first sorted chronologically according to trading date. Duplicate records, numerical fields, missing observations resulting from non-trading days, holidays, and stock-specific trading suspensions will be handled through chronological alignment of stock and NEPSE data. Prices and volume of all scrips and NEPSE index will be standardized into a common OHLCV structures with daily logarithmic returns computed from the adjusted closing prices.

3.2.2 Feature Engineering

A set of statistical and technical indicators will be generated from the OHLCV series to represent different dimensions of market behavior. The engineered features include:

Return Features: One-day return, Five-day return, Logarithmic return, Squared logarithmic return

Volatility Features: 10-day rolling volatility, 20-day rolling volatility, True Range (TR), Average True Range (ATR)

Momentum and Trend Features: Relative Strength Index (RSI), Average Directional Index (ADX), MACD Histogram, Local trend indicator derived from exponential moving averages

Volume Features: Normalized volume, On-Balance Volume (OBV) change

Market Interaction Features: Rolling 30-day correlation with NEPSE returns, Relative return against NEPSE

In addition to stock-specific variables, market-level features will be extracted from the NEPSE index, including NEPSE return, five-day return, rolling volatility, and RSI. These variables provide contextual information regarding overall market conditions.

3.2.3 Target Construction

The prediction task is formulated as a three-class classification. Let the 5-day forward return be defined as:

$$R_{t,t+5} = \frac{C_{t+5}}{C_t} - 1$$

The target label y_t is then assigned as follows:

$$y_t = \begin{cases} 2, & if\ R_{t,t+5} > 0.03 \\ 1, & if -0.03 \leq R_{t,t+5} \leq 0.03 \\ 0, & if\ R_{t,t+5} < -0.03 \end{cases}$$

where, 2 = Buy, 1 = Hold, 0 = Sell

This target construction is independent of the HMM states and therefore avoids circular labeling and avoids introducing regime-based information into the target construction process.

3.2.4 Chronological Data Partitioning

A time-based split avoids leakage. The dataset will be divided chronologically into train, validation, and test segments (70:15:15). Both the market-level HMM and the per-scrip HMMs, and prediction models will be fitted only on training observations, while test data will be reserved for final out-of-sample evaluation.

3.3 Implementation Model

3.3.1 Gaussian Hidden Markov Model for Regime Detection

This study will adopt a Gaussian Hidden Markov Model to capture the hidden dynamics of the financial markets exhibiting abrupt shifts between different volatility conditions, that are latent (unobserved) states, and which cannot be measured directly from raw price data.

An HMM assumes that an observed sequence is generated by an underlying sequence of hidden states. The hidden state at a particular time depends probabilistically on the state at the preceding time, while the observed variables depend on the current hidden state. A Gaussian Hidden Markov Model is a type of HMM where the hidden states generate continuous observations following a Gaussian (Normal) distribution. While a standard HMM often uses discrete symbols (like "Red" or "Blue"), a Gaussian HMM is essential for modeling real-world signals like audio waves, stock prices, or sensor readings.

A. Gaussian HMM Architecture

The Gaussian HMM used in this study consists of three main components.

1. A finite set of hidden volatility states

2. Transition probabilities controlling movement between states
3. State dependent Gaussian emission distributions generating financial features.

Let the hidden state at time (t) be denoted by

$$z_t \in \{1,2,\dots,K\}$$

where (K) represents the number of hidden states. In the proposed framework, three hidden states will be used and interpreted as:

$$S = \{LowVol, MidVol, HighVol\}$$

low volatility, medium volatility, and high volatility. The internal numerical state identifiers generated by the model, such as state 0, state 1, and state 2, do not have fixed financial meanings. They will therefore be relabeled after model estimation according to the average volatility of the observations assigned to each state. The general architecture of the 3-state Gaussian HMM is illustrated in Figure 3.2.

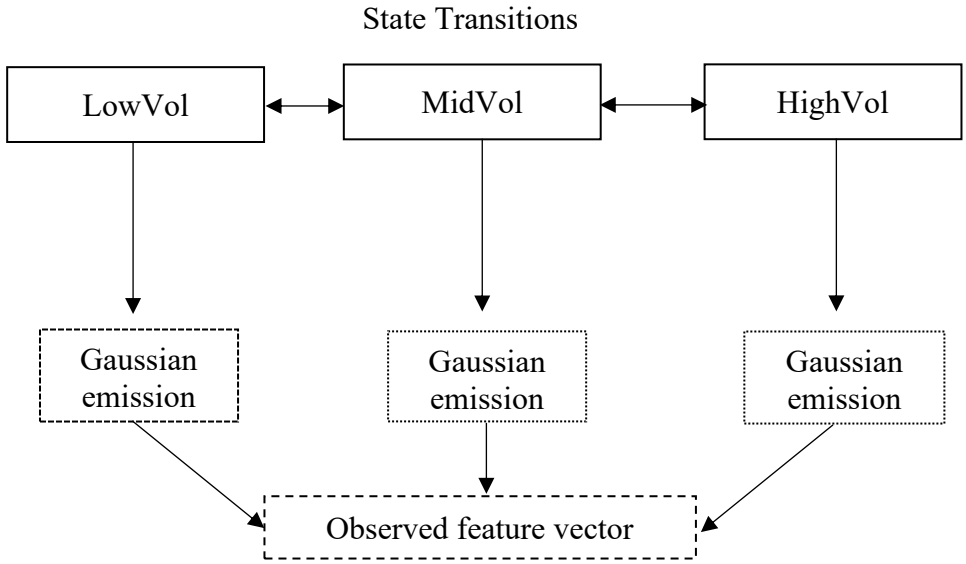


Figure 3.2: 3-state Gaussian HMM Architecture

A Gaussian HMM is represented by parameter set

$$\lambda = (\pi, A, \mu_k, \sum_k)$$

where, π represents initial state probability distribution, A denotes state transition probability matrix, μ_k is the mean vector of state k and, $\sum_k$ is the covariance matrix of state k.

The initial probability of starting in state (i) is expressed as:

The initial state distribution $\pi = [\pi_1, \pi_2, \dots, \pi_k]^T$ defines the probability of the system originating in state *i* at the initial time step *t = 1*:

$$\pi_i = P(z_1 = i), \text{ subject to } \sum_{i=1}^{K} \pi_i = 1$$

The dynamics of the hidden state sequence are governed by a first-order Markov process. The transition probability from state *i* at time *t* to state *j* at time *t+1* is defined conditionally as:

$$a_{ij} = P(z_{t+1} = j | z_t = i)$$

The full state transition behavior is collected into a stochastic matrix $\mathrm{A} \in \mathbb{R}^{KXK}$:

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1K} \\ \vdots & \ddots & \vdots \\ a_{K1} & \cdots & a_{KK} \end{bmatrix}$$

where each row forms a valid probability distribution such that:

$$\sum_{j=1}^{K} a_{ij} = 1, \forall \in \{1,2, \dots, K\}$$

The multi-feature observation vector at time *t* is denoted by the column vector $X_t \in \mathbb{R}^d$

$$Xt = [x_{t1}, x_{t2}, \dots, x_{td}]^T$$

where *d* represents the continuous feature dimensionality. Conditioned on the contemporary latent state $z_t = k$, the observation vector manifests as a multivariate Gaussian emission:

$$x_t \mid z_t = k \sim \mathcal{N}(\mu_k, \Sigma_k)$$

The associated state-dependent emission probability density function is parameterized by the mean vector $\mu_k \in \mathbb{R}^d$ and positive-definite covariance matrix$\Sigma_k \in \mathbb{R}^{dXd}$:

$$p(x_t \mid z_t = k) = \frac{1}{(2\pi)^{d/2}|\Sigma_k|^{1/2}} \exp\left[-\frac{1}{2}(x_t - \mu_k)^T \sum_k^{-1} (x_t - \mu_k)\right]$$

In this study, a diagonal covariance structure will be used. The diagonal structure allows each feature to have a different state-dependent variance while avoiding estimation of cross-covariance terms. The model parameters will be estimated using the Baum–Welch algorithm, which is an Expectation-Maximization procedure for HMMs

(Rabiner, 1989). In the implementation, this procedure is performed internally when the fit() method of the hmmlearn Gaussian HMM class is called. The selection of number of states will be assessed not only with Bayesian Information Criterion but also with financial interpretability and state occupancy. Hence, the proposed framework will use three states because they provide a practical representation of low, medium and high volatility conditions without excessive fragmentation of the time series.

B. Market-level and Stock-level Gaussian HMM

The first GHMM will model the overall NEPSE market. Its observation vector will consist of:

$$x_t^{(m)} = \begin{bmatrix} r_t^{NEPSE} \\ \sigma_t^{NEPSE} \\ RSI_{14,t}^{NEPSE} \end{bmatrix}$$

where, r_t^{NEPSE} denotes the daily log return of NEPSE index; σ_t^{NEPSE} is the 20-day rolling volatility of NEPSE returns, and $RSI_{14,t}^{NEPSE}$ is the 14-day NEPSE Relative Strength Index (RSI).

The market-level GHMM will generate one market regime for each available NEPSE trading date. The regime will then be attached to each stock observation having the same trading date. When a particular stock does not have an observation on that date, no artificial stock record will be created.

The second GHMM will be trained for each of the 18 selected scrips. The observation vector for stock (s) will be defined as:

$$x_t^{(s)} = \begin{bmatrix} r_t^s \\ \sigma_{20,t}^s \\ \rho_{30,t}^{s,m} \end{bmatrix}$$

Where, r_t^s denotes the daily log return of stock s, $\sigma_{20,t}^s$ is its 20-day rolling volatility, and $\rho_{30,t}^{s,m}$ is its trailing 30-day correlation with the NEPSE index return.

Separate estimation allows each scrip to have its own transition probabilities, Gaussian means, and variances. Since the numerical state identifiers produced by separately fitted models are arbitrary, the states of each stock will be ranked using their training-period average volatility and relabeled as *LowVol, MidVol,* and *HighVol*.

The Gaussian HMM component will produce two categorical variables: Market regime and Stock regime. These variables will be represented by the labels LowVol, MidVol, and HighVol. They will be combined with feature engineered technical, volatility, and market-related features and supplied to the LightGBM and XGBoost classifiers.

3.3.2 Gradient Boosting Models for Prediction

This study will use and compare two gradient boosting models: XGBoost and LightGBM. These models are suitable for the proposed framework because the input data are represented as structured tabular features and include both numerical and categorical variables such as engineered technical indicators, market-related features, scrip identifier, market regime, and stock regime.

Gradient boosting is an ensemble learning approach that combines multiple weak learners, usually decision trees, into a strong predictive model. The model is built sequentially, where each new tree is trained to reduce the error made by the previous trees.

For an input feature vector $\mathrm{X} \in \mathbb{R}^{D}$, the additive ensemble model generated via gradient boosting at iteration *M* is formalized as:

$$F_M(X) = \sum_{m=1}^{M} \eta f_m(x)$$

where $M \in \mathrm{Z}^{+}$ represents the total number of boosting iterations, $f_m(x)$denotes the base learner appended to the ensemble at iteration m, and $\eta \in (0,1)$ is the shrinkage parameter or learning rate governing the regularization of each base learner's contribution.

For multi-class classification, separate score functions are estimated for each class. The probability of class (k) is obtained using the SoftMax function:

$$P(y = k|X) = \frac{\exp\left(F_{M,k}(x)\right)}{\sum_{c=1}^{C} \exp\left(F_{M,c}(x)\right)}$$

Where (C=3) for the Buy, Hold, Sell classes. The final prediction class is

$$\hat{y} = argmax_k P(y = k|x)$$

The nominal target labels within the sample space are encoded using an ordinal numerical mapping: Sell:0, Hold:1, Buy: 2

The prediction models will be trained using a stacked panel dataset, where observations from all selected scrips are combined. The scrip identifier will be included as a categorical feature so that the model can learn both shared market patterns and stock-specific behavior.

A. LightGBM Model

LightGBM is a gradient boosting framework that uses tree-based learning algorithms. It is designed for speed and lower memory usage, specifically for high-dimensional and large-scale data. It uses a leaf-wise (vertical) tree growth strategy. LightGBM grows tree leaf-wise. It chooses the leaf with the maximum loss reduction to split, regardless of the level (Ke, et al., 2017).

In this study, LightGBM will be used as one of the main classifiers for Buy, Hold, and Sell prediction. It will receive engineered stock features, market features, and HMM-based regime features as inputs. The model output will be a probability distribution over the three classes, and the class with the highest probability will be selected as the final prediction. The LightGBM architecture used in the proposed framework can be summarized as:

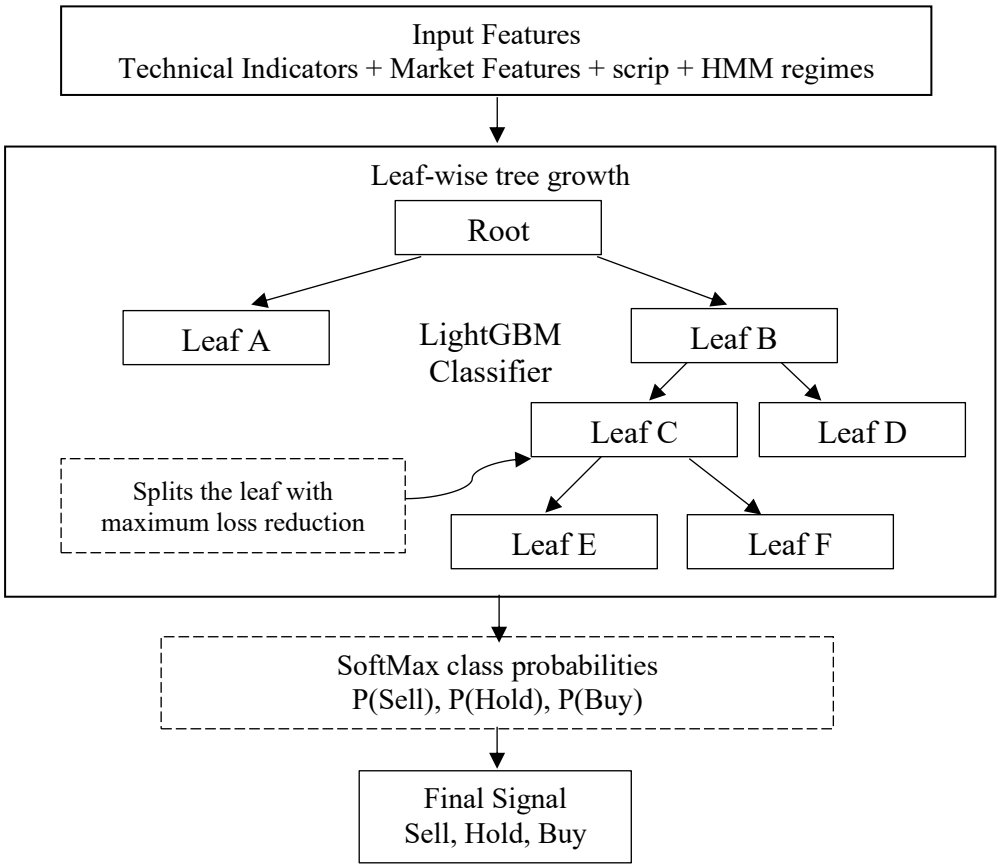


Figure 3.3: LightGBM Architecture

B. XGBoost Model

XGBoost is an optimized distributed gradient boosting library designed to be highly efficient, flexible, and portable. It implements machine learning algorithms under the Gradient Boosting framework. It uses a level-wise (horizontal) tree growth strategy. XGBoost grows trees level-by-level. It scans all possible split points across all features to find the one that maximizes the gain. The XGBoost architecture used in the proposed framework can be summarized as:

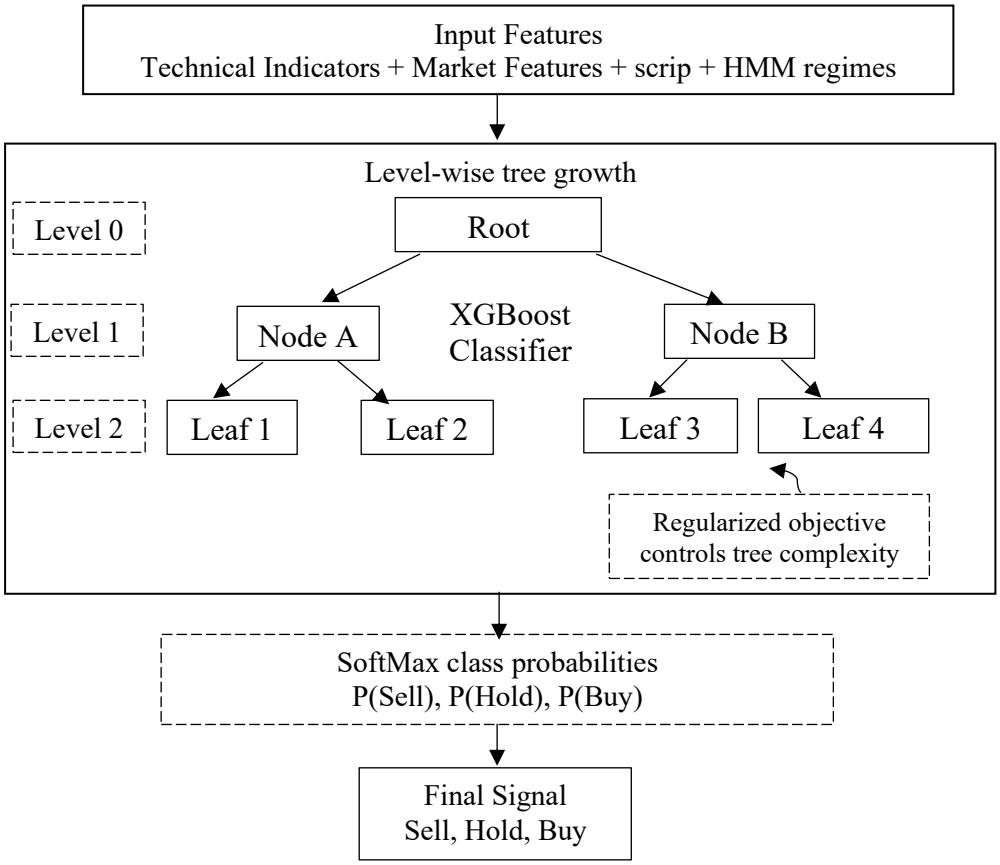


Figure 3.4: XGBoost Architecture

At boosting iteration *M*, the objective function minimized by the regularized gradient boosting framework is defined as:

$$\mathcal{L}^{M} = \sum_{i=1}^{n} l(y_i, \widehat{y_i^{m}}) + \sum_{m=1}^{M} \Omega(f_m)$$

Where, $l(y_i, \widehat{y_i^{m}})$ represents a differentiable training loss function measuring the discrepancy between the target label y_i and the predicted value $\widehat{y_i^{m}}$, and $\Omega(f_m)$ is the structural regularization penalty assigned to the m-th tree learner f_m. This regularization term is parameterized as:

$$\Omega(f_{\{m\}}) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2$$

Where, (T) is the number of leaves in the tree; w_j is the score assigned to leaf (j); γcontrols the penalty for adding leaves; and λ controls the penalty on leaf weights.

This regularized objective helps XGBoost control model complexity and improve generalization (Chen & Guestrin, 2016).

In this study, to evaluate whether regime information improves prediction, both models will be trained on the stacked panel dataset under four feature configurations: baseline, baseline + market regime, baseline + stock regime, and baseline + both regimes, and the best model will be chosen based on the evaluation metrics.

The baseline model will allow comparison against a feature-only approach, while the regime-aware models will test whether HMM-derived volatility states provide additional predictive value.

C. Training Procedure

The model training procedure will follow a leakage-aware chronological design. First, the combined panel dataset will be divided into training, validation, and test periods based on time order rather than random sampling. Feature engineering and regime variables will be prepared using only past and training period information to prevent look-ahead bias. LightGBM and XGBoost models will be then trained on the training set, while the validation set will be used for model selection and hyperparameter adjustment. Finally, the test set will be reserved for out-of-sample evaluation, and the best-performing model will be selected using predictive metrics, with main focus on Macro F1-score.

Through this design, the study will compare whether adding market-level and stock-level HMM regimes improve the predictive performance of gradient boosting models for NEPSE stock movement prediction.

3.4 Evaluation Metrics

To evaluate the effectiveness of the proposed models in detecting the stock market signals, both classification task metrics and trading-oriented metrics will be employed highlighting the need to correctness and practical usefulness.

3.4.1 Macro F1 Score

Since the classes Buy, Hold, and Sell are not perfectly balanced, the primary predictive metric is the Macro F1-score. The F1-score for a class is defined as:

$$\mathrm{F1} = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

The macro F1-score is the unweighted average of class-wise F1-scores:

$$F1_{macro} = \frac{1}{C}\sum_{c=1}^{C} F1_c$$

where C is the number of classes. This metric is important because it evaluates the model's ability to perform reasonably across all classes rather than being dominated by the majority class.

3.4.2 Cumulative Return

Cumulative return over the backtest period is defined as:

$$CR = \prod_{t=1}(1 + r_t) - 1$$

where r_t is the return earned during the t-th trade period or portfolio period. Cumulative returns summarize the total growth of the strategy over time.

3.4.3 Sharpe Ratio

Sharpe ratio measures risk-adjusted return:

$$Sharpe = \frac{\bar{r} - rf}{\sigma r}$$

where $\bar{r}$ is the average strategy return, r_f is the risk-free rate, and σ_r is the standard deviation of returns. In practical implementation, the risk-free rate may be treated as zero for simplicity if a short-horizon comparison is used. Sharpe ratio is one of the most important trading metrics in this study because it evaluates return relative to volatility.

3.4.4 Maximum Drawdown

Maximum drawdown measures the largest decline from a historical peak in the equity curve:

$$MDD = min_t\left(\frac{Vt - max_{r\leq t}Vt}{max_{r\leq t}Vt}\right)$$

where V_t is portfolio value at time t. MDD is important because it captures downside risk and helps compare the strategy's loss severity with that of a benchmark.

3.4.5 Win Rate

Win rate is defined as the percentage of trades with positive realized return:

$$\text{Win Rate} = \frac{Number\ of\ Profitable\ Trades}{Total\ Number\ of\ Trades}$$

Although win rate is not sufficient by itself, it provides an intuitive measure of how often the strategy produces successful trades.

CHAPTER 4: EXPECTED OUTCOMES AND WORK SCHEDULE

4.1 Expected Outcomes

This study is expected to produce a regime-aware stock movement prediction framework for NEPSE that combines HMMs for volatility regime detection with gradient boosting models for Buy, Hold, and Sell classification. The research is expected to identify meaningful market-level and stock-level volatility regimes, determine whether regime information improves predictive performance over baseline feature-based models, and provide empirical evidence on the relative usefulness of LightGBM and XGBoost in the NEPSE context. In addition, the study is expected to generate a leakage-aware evaluation pipeline and a simplified out-of-sample backtesting framework that will help assess whether the predicted signals are economically useful. Overall, the expected outcome is both methodological and practical: a structured regime-aware prediction approach tailored to an under-researched emerging market, together with evidence on how regime information affects stock movement prediction and trading signal quality in NEPSE.

4.2 Work Schedule

The Gantt Chart below depicts the working schedule:

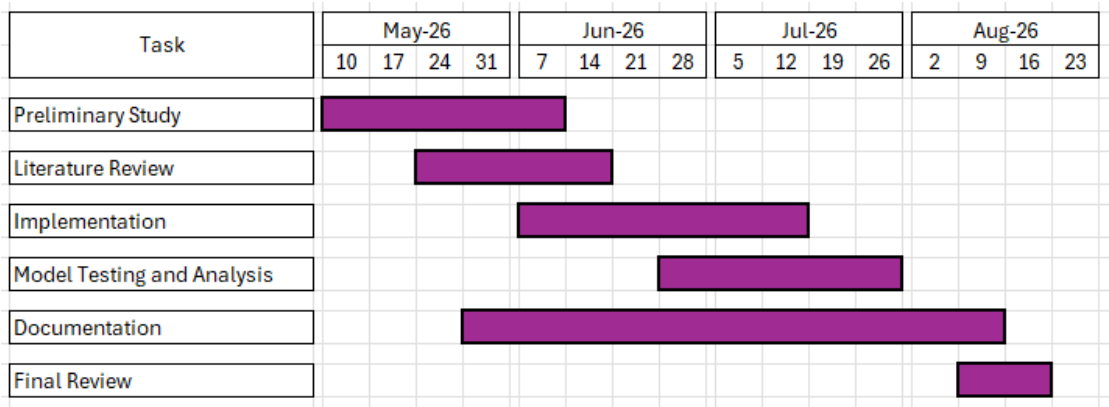


Figure 4.1: Working Schedule

REFERENCES

Bhusal, M. K. (2017). Application of Markov Chain Model in the Stock Market Trend Analysis of Nepal . *International Journal of Scientific & Engineering Research, 8*(10).

Bisdoulis, K. -L. (2025). Assets forecasting with feature engineering and transformation methods for LightGBM. *arXiv*. From https://doi.org/10.48550/arXiv.2501.07580

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proc. 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785-794*.

Dahal, K. R., Gupta, A., & Pokharel, N. R. (2024). Predicting the Direction of NEPSE Index Movement with News Headlines Using Machine Learning. *Econometrics, 12*(2), 16.

Fons, E., Dawson, P., Yau, J., Zeng, X.-J., & Keane, J. (2021). A novel dynamic asset allocation system using Feature Saliency Hidden Markov models for smart beta investing. *Expert Systems with Applications, 163*, 113720. doi:10.1016/j.eswa.2020.113720

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems, 30*.

Kijkarncharoensin, A. (2025). Identifying Risk Regimes in a Sectoral Stock Index Through a Multivariate Hidden Markov Framework. *Risks, 13*(7), 135.

Koukorinis, A., Germano, G., & Peters, G. W. (2025). Generative-Discriminative Machine Learning Models for High-Frequency Financial Regime Classification. *Methodol Comput Appl Probab, 27*(36). From https://doi.org/10.1007/s11009-025-10148-8

Okşak, Y., Büyükkör, Y., & Sarıtaş, T. (2026). Wavelet-enhanced multimodel framework for stock market forecasting: A comprehensive analysis across market regimes. *Borsa Istanbul Review, 26*(1). doi:https://doi.org/10.1016/j.bir.2025.100771.

Olorunnimbe, K., & Viktor, H. (2023). Deep learning in the stock market—a systematic survey of practice, backtesting, and applications. *Artificial Intelligence Review*. doi:10.1007/s10462-022-10226-0

Pabuccu, H., & Barbu, A. (2024). Feature selection with annealing for forecasting financial time series. *Financial Innovation, 10*, Article 87. doi:https://doi.org/10.1186/s40854-024-00617-3

Pagliaro, A. (2026). Regime-Aware LightGBM for Stock Market Forecasting: A Validated Walk-Forward Framework with Statistical Rigor and Explainable AI Analysis. *Electronics, 15(6)*(1334). From https://doi.org/10.3390/electronics15061334

Pokharel, N. R., Dahal, K. R., Rimal, R., Bhandari, H., K.C. Khatri, R., Rimal, B., & Hahn, W. E. (2022). Predicting NEPSE index price using deep learning models. *Machine Learning with Applications, 9*.

Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE, 77*(2), 257-286. doi:10.1109/5.18626

ShareSansar. (n.d.). *www.sharesansar.com*. Retrieved January 24, 2026 from https://www.sharesansar.com/company/{symbol};
https://www.sharesansar.com/index-history-data

Thakkar, A., & Chaudhari, K. (2021). A comprehensive survey on deep neural networks for stock market: The need, challenges, and future directions. *Expert Systems with Applications, 177*, 114800. doi:10.1016/j.eswa.2021.114800